

2 From Critical Height to Compatible Finite Views

2.1 Critical Height at a Finite Endpoint

Fix a rapidly decaying datum u_0 on $\mathbb{R}^3$, a viscosity $\nu > 0$, and the maximal classical solution (u, p) they generate on $[0, T_*)$. Assume $T_* < \infty$. The dilation and the decaying pressure normalization used below are whole-space statements. On $\mathbb{T}^3$, discrete frequencies replace this dilation and require a separate high-frequency or localization argument.

The solution is smooth at every earlier time, so any singular behaviour must be approached through a sequence of regular states as $t \uparrow T_*$. Kinetic energy controls the total L^2 size of those states. That L^2 ceiling remains compatible with rescaled states at progressively shorter length scales. Navier–Stokes scaling singles out the Sobolev height that stays unchanged under exactly that compression.

If (u, p) solves the Navier–Stokes equations with viscosity $\nu > 0$, then for every $\lambda > 0$,

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

solves the same equation with the same viscosity on the rescaled time interval. The spatial scale is divided by λ , while the corresponding time scale is divided by λ^2 . This is the parabolic relation already present in $\partial_t - \nu \Delta$.

Let $\Lambda = (-\Delta)^{1/2}$, and write

$$\|v\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{v}(\xi)|^2 d\xi.$$

The Fourier transform of the rescaled velocity is

$$\widehat{u}_\lambda(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t).$$

Substituting this expression and setting $\eta = \xi/\lambda$ gives

$$\begin{aligned} \|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} \left| \widehat{u}(\xi/\lambda, \lambda^2 t) \right|^2 d\xi \\ &= \lambda^{2s-1} \int_{\mathbb{R}^3} |\eta|^{2s} \left| \widehat{u}(\eta, \lambda^2 t) \right|^2 d\eta. \end{aligned}$$

Hence

$$\|u_\lambda(t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\lambda^2 t)\|_{\dot{H}^s}.$$

The exponent vanishes at $s = 1/2$. This gives the critical height

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2.$$

The scaling can compress a smooth configuration to a finer spatial length and a shorter time while preserving the value of H .

At the kinetic-energy level, the same calculation gives

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2.$$

Successive rescalings can therefore lower the kinetic-energy scale while leaving the critical height unchanged. This is the precise opening through which a finite-energy estimate can fail to see a concentration at ever finer scales.

The finite endpoint gives the scaling a second consequence. Choose a geometric sequence of lengths

$$\ell_k := 2^{-k} \ell_0.$$

The viscous time associated with ℓ_k has parabolic size

$$\tau_k := \frac{c\ell_k^2}{\nu} = \frac{c\ell_0^2}{\nu} 4^{-k},$$

where $c > 0$ fixes the chosen normalization. These times have a finite sum,

$$\sum_{k=0}^{\infty} \tau_k = \frac{4c\ell_0^2}{3\nu} < \infty.$$

Parabolic timing therefore permits an abstract Zeno schedule in which infinitely many progressively finer episodes fit before a finite time. The calculation establishes a dimensionally consistent schedule. Realizing that schedule would require the same Navier–Stokes history to generate every episode while transport, pressure, incompressibility, and viscosity continue to act together.

The critical height tells us which spatial measurement can remain visible through such a schedule. The speed at which the field moves from one regular state to the next is still unknown. Finite time makes that response question exact: an unbounded rung can be reached only through the accumulated action of the rung above it.

2.2 Finite-Time Response Escalation

Set

$$V_j(t) := \partial_t^j u(\cdot, t), \quad j \in \mathbb{N}_0.$$

These are Eulerian time derivatives: the spatial coordinate is held fixed while time changes. Let X be any Banach space in which two consecutive rows are defined on a strict interval $[t_0, t] \subset [0, T_*)$. The fundamental theorem of calculus gives

$$V_j(t) = V_j(t_0) + \int_{t_0}^t V_{j+1}(\tau) d\tau.$$

Taking the X -norm gives

$$\|V_j(t)\|_X \leq \|V_j(t_0)\|_X + \int_{t_0}^t \|V_{j+1}(\tau)\|_X d\tau.$$

Thus

$$\int_{t_0}^{T_*} \|V_{j+1}(\tau)\|_X d\tau < \infty$$

gives a finite upper bound for V_j throughout the remaining interval. It also gives a unique endpoint value in X , since for $t_0 < s < t < T_*$,

$$\|V_j(t) - V_j(s)\|_X \leq \int_s^t \|V_{j+1}(\tau)\|_X d\tau.$$

The absolute continuity of the integral makes $V_j(t)$ Cauchy as $t \uparrow T_*$.

Now suppose that the j -th row becomes unbounded in X :

$$\sup_{t_0 \leq t < T_*} \|V_j(t)\|_X = \infty.$$

The preceding estimate forces

$$\int_{t_0}^{T_*} \|V_{j+1}(\tau)\|_X d\tau = \infty.$$

Since $T_* - t_0$ is finite,

$$\int_{t_0}^{T_*} \|V_{j+1}(\tau)\|_X d\tau \leq (T_* - t_0) \sup_{t_0 \leq t < T_*} \|V_{j+1}(t)\|_X$$

whenever the supremum on the right is finite. Consequently,

$$\sup_{t_0 \leq t < T_*} \|V_j(t)\|_X = \infty \implies \sup_{t_0 \leq t < T_*} \|V_{j+1}(t)\|_X = \infty.$$

Repeating the same argument gives

$$\sup_{t_0 \leq t < T_*} \|V_{j+k}(t)\|_X = \infty \quad \text{for every finite } k \geq 1.$$

This is the exact content of the earlier statement that a finite-time blow-up must be led by faster responses above it. The statement begins at whichever Eulerian row first becomes unbounded in the chosen norm. Velocity can remain bounded while a gradient or a higher time derivative becomes the first escaping quantity; the same argument then starts at that coordinate. The implication concerns size alone. Each higher row may oscillate, change direction, and pass through zero while still having the unbounded size required by the implication.

The physical language of acceleration and jerk follows a moving fluid particle, whereas V_1 and V_2 hold space fixed. The distinction matters because transport separates these two kinds of differentiation. The original equation tells us exactly how they fit together, and differentiating that equation produces the full response family forced by the fluid itself.

2.3 The Pressure-Complete Mixed Jet

Let $X(a, t)$ be the path of the particle that starts at a , so

$$\frac{d}{dt}X(a, t) = u(X(a, t), t).$$

Along this path,

$$\frac{d}{dt}u(X(a, t), t) = \partial_t u + (u \cdot \nabla)u =: D_t u.$$

The material derivative $D_t u$ is the particle's acceleration. The Navier–Stokes equation gives its complete value,

$$D_t u = -\nabla p + \nu \Delta u.$$

One more carried derivative gives

$$D_t^2 u = \partial_t^2 u + 2(u \cdot \nabla) \partial_t u + (\partial_t u \cdot \nabla) u + (u \cdot \nabla)((u \cdot \nabla) u).$$

The material jerk is therefore built from Eulerian time derivatives, spatial derivatives, and products of lower responses. The Eulerian family $V_n = \partial_t^n u$ gives fixed-coordinate analytic coordinates for these carried responses.

Return to the original system and set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

The undifferentiated row is

$$V_1 + (V_0 \cdot \nabla) V_0 + \nabla Q_0 = \nu \Delta V_0, \quad \nabla \cdot V_0 = 0.$$

Differentiate once in time. Leibniz's rule gives

$$\partial_t((V_0 \cdot \nabla) V_0) = (V_1 \cdot \nabla) V_0 + (V_0 \cdot \nabla) V_1,$$

and hence

$$V_2 + (V_1 \cdot \nabla) V_0 + (V_0 \cdot \nabla) V_1 + \nabla Q_1 = \nu \Delta V_1.$$

One more differentiation gives

$$V_3 + (V_2 \cdot \nabla) V_0 + 2(V_1 \cdot \nabla) V_1 + (V_0 \cdot \nabla) V_2 + \nabla Q_2 = \nu \Delta V_2.$$

The coefficients 1, 2, 1 appear because two time derivatives can be divided between the transporting and transported factors in those three ways.

At temporal order n , the same product rule gives

$$\partial_t^n((u \cdot \nabla) u) = \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a}.$$

The full temporal recurrence is therefore

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n, \quad \nabla \cdot V_n = 0.$$

The new time row V_{n+1} is joined to every lower temporal split of transport, to the pressure response at the same depth, and to two additional spatial derivatives through viscosity.

Pressure is fixed by the same velocity rows. Taking the divergence of the undifferentiated equation and using $\nabla \cdot u = 0$ gives

$$-\Delta p = \partial_i \partial_j (u_i u_j).$$

After n time derivatives,

$$-\Delta Q_n = \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

Rapid decay fixes the additive normalization, equivalently

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

Thus every temporal pressure row is obtained from the velocity products in the same temporal row.

The recurrence remains open in spatial order, so differentiate it as well. For a multi-index $\alpha \in \mathbb{N}_0^3$, define

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

For $\beta \leq \alpha$, write

$$\binom{\alpha}{\beta} := \prod_{k=1}^3 \binom{\alpha_k}{\beta_k}.$$

The spatial product rule applied to each temporal split gives

$$\partial_t^n \partial_x^\alpha ((u \cdot \nabla) u) = \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta}.$$

Substitution gives the pressure-complete mixed recurrence

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} \\ + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0, \end{aligned}$$

with

$$-\Delta \Pi_{n,\alpha} = \partial_i \partial_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta})_i (J_{n-a,\alpha-\beta})_j.$$

Every coordinate in this family is a derivative of the same (u, p) , and every row retains the transport, viscous, divergence, and pressure relations that produced it. The mixed recurrence gives the full derivative family. The critical height now gives a way to see, row by row, how its temporal direction contains an ascending spatial Sobolev column.

2.4 The Spatial Column Inside Critical Height

For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2.$$

The critical height is $H = E_{1/2}$. Its derivatives must retain the terms that drive it, so first derive the balance at an arbitrary spatial height s .

Apply Λ^s to the momentum equation and take the L^2 inner product with $\Lambda^s u$. The time derivative gives

$$\langle \Lambda^s u, \Lambda^s \partial_t u \rangle = \frac{d}{dt} \frac{1}{2} \|\Lambda^s u\|_2^2 = E'_s.$$

Rapid decay and the commutation of Λ^s with spatial derivatives give

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle = -\nu \left\| \Lambda^{s+1} u \right\|_2^2 = -2\nu E_{s+1}.$$

The pressure pairing vanishes in this global balance:

$$\langle \Lambda^s u, \Lambda^s \nabla p \rangle = -\langle \nabla \cdot \Lambda^s u, \Lambda^s p \rangle = 0.$$

For the remaining signed transfer, write

$$\mathcal{P}_s(t) := -\langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle.$$

The pressure term is retained in this definition because $\mathcal{P}_s$ comes from the complete momentum equation, even though its global pairing is zero. The exact balance is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

One time derivative of the spatial energy at height s therefore contains the energy one full Sobolev step higher. It also contains the signed transfer $\mathcal{P}_s$ generated by the lower rows of the mixed recurrence.

Put $s = 1/2$. The first critical-height row is

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

The next row requires the balance at $s = 3/2$,

$$E'_{3/2} = \mathcal{P}_{3/2} - 2\nu E_{5/2}.$$

Substituting it into the derivative of H' gives

$$\begin{aligned} H'' &= \mathcal{P}'_{1/2} - 2\nu E'_{3/2} \\ &= \mathcal{P}'_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

The third derivative uses

$$E'_{5/2} = \mathcal{P}_{5/2} - 2\nu E_{7/2}$$

and gives

$$H''' = \mathcal{P}''_{1/2} - 2\nu \mathcal{P}'_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

One more differentiation gives

$$H^{(4)} = \mathcal{P}'''_{1/2} - 2\nu \mathcal{P}''_{3/2} + (2\nu)^2 \mathcal{P}'_{5/2} - (2\nu)^3 \mathcal{P}_{7/2} + (2\nu)^4 E_{9/2}.$$

The highest spatial energy advances by one Sobolev step each time, while every transfer encountered on the way remains in the row.

The pattern follows by induction. For every integer $n \geq 1$,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

Indeed, differentiating the n -th row differentiates each existing transfer term. Replacing

$$E'_{n+1/2} = \mathcal{P}_{n+1/2} - 2\nu E_{n+3/2}$$

creates the next transfer term and the next highest spatial energy with exactly the displayed coefficient.

The first rows can now be read with their distinct quantities intact:

temporal row	highest spatial energy appearing in that complete row
H	$E_{1/2}$
H'	$-2\nu E_{3/2}$
H''	$(2\nu)^2 E_{5/2}$
H'''	$-(2\nu)^3 E_{7/2}$
$H^{(4)}$	$(2\nu)^4 E_{9/2}$

For example, H' is the full quantity $\mathcal{P}_{1/2} - 2\nu E_{3/2}$. The table identifies the highest spatial term inside each complete temporal row.

The energies E_s use the homogeneous Sobolev scale because that scale makes the dilation exact. Continuation is usually stated in the inhomogeneous scale. For every $s > 0$,

$$\|u(t)\|_{H^s}^2 \simeq \|u(t)\|_2^2 + \|\Lambda^s u(t)\|_2^2 = \|u(t)\|_2^2 + 2E_s(t).$$

The unforced energy identity gives

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2,$$

so the low-frequency term is already controlled by the datum. A bound for E_s therefore supplies the corresponding inhomogeneous H^s bound, and an L^1 -in-time bound for E_s supplies the high-frequency part of an $L_t^2 H_x^s$ bound on every finite interval.

The critical-height identities reveal the spatial heights present in the temporal tower. Their result at this stage is structural because the signed transfer terms can cancel the highest spatial term within a row. Following any finite number of temporal rows reaches one finite spatial height. Smoothness asks for every finite height, which turns the discovery into an intersection question.

2.5 Simultaneous Sobolev Intersections

The spatial column becomes smooth one requested derivative at a time. In three dimensions,

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3) \quad \text{whenever } m > k + \frac{3}{2}.$$

Fix one derivative order k . One finite Sobolev height $m > k + 3/2$ controls every spatial derivative through order k . Since the choice can be made for every finite k ,

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Each requested derivative uses one finite row high enough for that request.

The temporal column works in the same coordinatewise way on a strict finite interval $I = (0, T)$, $T < T_*$. For a Hilbert space X ,

$$H^r(I; X) \hookrightarrow C^q(\bar{I}; X) \quad \text{whenever } r > q + \frac{1}{2}.$$

Fixing q and choosing one finite $r > q + 1/2$ gives the desired temporal derivative. Thus

$$\bigcap_{r=0}^{\infty} H^r(I; X) \hookrightarrow C^\infty(\bar{I}; X).$$

Applied to the scalar critical height, this says

$$H \in \bigcap_{r=0}^{\infty} H^r(I) \implies H \in C^\infty(\bar{I}).$$

This scalar intersection describes the complete time regularity of H . The velocity field also carries spatial direction, phase, and geometry, all of which lie outside the scalar value $H(t)$.

The velocity therefore needs both indices. Define its coordinatewise mixed intersection by

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)).$$

The meaning of the intersection becomes concrete when one mixed derivative is requested. For example, continuity of

$$\partial_t^2 \partial_x^\alpha u, \quad |\alpha| \leq 4,$$

follows from the single finite membership

$$u \in H^3(I; H^6(\mathbb{R}^3)),$$

because $3 > 2 + 1/2$ and $6 > 4 + 3/2$. In general, for any finite temporal order q and spatial order k , choose finite indices

$$r > q + \frac{1}{2}, \quad m > k + \frac{3}{2}.$$

Then

$$H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Arbitrary finite q and k give

$$\mathcal{V}_\infty(I) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3).$$

Membership is asserted one finite coordinate at a time:

$$u \in \mathcal{V}_\infty(I) \iff \|u\|_{H^r(I; H^m)} < \infty \quad \text{for every finite } (r, m).$$

The bound may change with r and m . The intersection places all of those finite statements on the same function u .

This is the velocity intersection. The pressure-complete Navier–Stokes object also contains $Q_n = \partial_t^n p$, the differentiated momentum equation, the divergence constraint, and the pressure Poisson equation at every retained depth. Those relations were derived in the mixed recurrence. The next step is to find a finite piece of the velocity intersection that produces an actual time trace while carrying those pressure-complete relations with it.

2.6 Finite Adjacent Rectangles

Begin with the first finite case that carries two temporal rows to a common trace height. Fix a strict interval $I = (0, T)$, $T < T_*$, and write

$$V_0 = u, \quad V_1 = \partial_t u, \quad V_2 = \partial_t^2 u.$$

Suppose

$$V_0, V_1 \in L^2(I; H^3(\mathbb{R}^3))$$

and

$$V_1, V_2 \in L^2(I; H^1(\mathbb{R}^3)).$$

For V_0 , these assumptions give

$$V_0 \in L^2(I; H^3), \quad \partial_t V_0 = V_1 \in L^2(I; H^1).$$

The Hilbert triple

$$H^3(\mathbb{R}^3) \hookrightarrow H^2(\mathbb{R}^3) \hookrightarrow H^1(\mathbb{R}^3)$$

uses H^2 as the pivot space. For a smooth function v , the identity behind its trace is

$$\|v(t)\|_{H^2}^2 - \|v(s)\|_{H^2}^2 = 2 \int_s^t \langle \partial_\tau v(\tau), v(\tau) \rangle_{H^1, H^3; H^2} d\tau.$$

The right side is integrable when

$$v \in L^2(I; H^3), \quad \partial_t v \in L^2(I; H^1).$$

The standard density argument then gives $v \in C([0, T]; H^2)$. Applying this identity to V_0 gives

$$V_0 \in C([0, T]; H^2(\mathbb{R}^3)).$$

Applying the same argument to V_1 , using $\partial_t V_1 = V_2$, gives

$$V_1 \in C([0, T]; H^2(\mathbb{R}^3)).$$

The two adjacent spatial rows have produced time traces for both retained temporal coordinates.

Collect the four norms in

$$\begin{aligned} \mathfrak{R}_{1,2}(u; T) := & \|u\|_{L^2(0, T; H^3)}^2 + \|\partial_t u\|_{L^2(0, T; H^3)}^2 \\ & + \|\partial_t u\|_{L^2(0, T; H^1)}^2 + \|\partial_t^2 u\|_{L^2(0, T; H^1)}^2. \end{aligned}$$

Its coordinate pattern is

	$n = 0$	$n = 1$
H^3	u	$\partial_t u$
H^1	$\partial_t u$	$\partial_t^2 u$

The top and bottom entries in each column form one adjacent Hilbert-triple pair. The repeated $\partial_t u$ joins the first temporal block to the second. This finite two-by-two pattern is the first adjacent rectangle.

The same construction works at arbitrary finite depths. Fix independent integers $N, M \in \mathbb{N}_0$. The middle trace height is H^M , with adjacent spaces

$$H^{M+1}(\mathbb{R}^3) \hookrightarrow H^M(\mathbb{R}^3) \hookrightarrow H^{M-1}(\mathbb{R}^3).$$

For $M = 0$, the lower space is the inhomogeneous space $H^{-1}(\mathbb{R}^3)$. Define the finite velocity norm

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) &:= \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned}$$

Finiteness of this quantity gives, for each $0 \leq n \leq N$,

$$V_n \in L^2(0, T; H^{M+1}), \quad \partial_t V_n = V_{n+1} \in L^2(0, T; H^{M-1}).$$

For a smooth representative, these two rows meet in the exact calculation

$$\left| \frac{d}{dt} \|V_n(t)\|_{H^M}^2 \right| \leq 2 \|V_{n+1}(t)\|_{H^{M-1}} \|V_n(t)\|_{H^{M+1}}.$$

The right side belongs to $L^1(0, T)$ by Cauchy–Schwarz. Approximation by smooth functions, together with weak continuity, therefore gives the H^M -continuous representative. The same trace argument yields

$$V_n \in C([0, T]; H^M(\mathbb{R}^3)), \quad 0 \leq n \leq N.$$

Increasing N adds temporal rows. Increasing M raises the spatial trace height. The two choices remain independent.

The number $\mathfrak{R}_{N,M}(u; T)$ measures velocity rows only. The pressure-complete finite record must identify those rows as derivatives of the same Navier–Stokes history. For the fixed whole-space solution (u, p) generated by u_0 , define

$$\mathcal{R}_{N,M}[u_0; T] := \left((V_n)_{n=0}^{N+1}, (Q_n)_{n=0}^N; \mathfrak{R}_{N,M}(u; T) \right),$$

where

$$V_n = \partial_t^n u, \quad Q_n = \partial_t^n p,$$

the velocity rows have the memberships counted by $\mathfrak{R}_{N,M}$, and for every $0 \leq n \leq N$,

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n,$$

$$\nabla \cdot V_n = 0,$$

and

$$-\Delta Q_n = \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

The rapid-decay normalization fixes each Q_n . Every mixed coordinate $J_{n,\alpha}$ available at the retained Sobolev depths is the weak spatial derivative $\partial_x^\alpha V_n$, and its pressure coordinate is the corresponding derivative of Q_n . Thus $\mathfrak{R}_{N,M}$ is the finite velocity norm, while $\mathcal{R}_{N,M}$ is the finite pressure-complete Navier–Stokes object built around that norm.

Every finite rectangle supplies only the traces at its chosen temporal and spatial depths. The all-orders velocity intersection is recovered by allowing all finite pairs (N, M) . Those rectangles belong to one intersection when every shared coordinate agrees, which is the compatibility question.

2.7 Compatibility and the Terminal Quantifier

Take the concrete rectangle $\mathcal{R}_{2,3}[u_0; T]$. Its upper velocity row contains

$$V_0, V_1, V_2 \in L^2(0, T; H^4),$$

and its lower derivative row contains

$$V_1, V_2, V_3 \in L^2(0, T; H^2).$$

It also contains Q_0, Q_1, Q_2 with their three differentiated momentum, divergence, and pressure equations.

Now retain temporal orders through $n = 1$ and lower the middle spatial height from H^3 to H^2 . The embeddings

$$H^4 \hookrightarrow H^3, \quad H^2 \hookrightarrow H^1$$

give

$$V_0, V_1 \in L^2(0, T; H^3)$$

and

$$V_1, V_2 \in L^2(0, T; H^1).$$

Retaining Q_0, Q_1 and the corresponding equations gives exactly $\mathcal{R}_{1,2}[u_0; T]$. Every surviving entry is the same derivative it was in the larger rectangle. The operation changes the finite depth at which the history is viewed and preserves the history itself.

The general restriction has three independent cutoffs. Let

$$N' \leq N, \quad M' \leq M, \quad 0 < T' \leq T < T_*.$$

Define

$$\rho_{N',M',T'}^{N,M,T} : \mathcal{R}_{N,M}[u_0; T] \longrightarrow \mathcal{R}_{N',M'}[u_0; T']$$

by retaining $V_0, \dots, V_{N'+1}$, retaining $Q_0, \dots, Q_{N'}$, using the continuous embeddings from the M -spaces to the M' -spaces, and restricting every retained function to $(0, T')$. For every shared temporal and spatial index,

$$J_{n,\alpha} = \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} = \partial_t^n \partial_x^\alpha p$$

remain the same distributional derivatives. The retained rows continue to satisfy the same pressure-complete mixed recurrence.

Restriction in stages agrees with direct restriction. If

$$N'' \leq N' \leq N, \quad M'' \leq M' \leq M, \quad T'' \leq T' \leq T,$$

then

$$\rho_{N'', M'', T''}^{N', M', T'} \circ \rho_{N', M', T'}^{N, M, T} = \rho_{N'', M'', T''}^{N, M, T}.$$

This is compatibility. The same lower derivative is obtained from every larger rectangle that contains it, regardless of the intermediate cutoffs. The coordinatewise intersections from the previous subsection therefore refer to one velocity–pressure history.

One issue remains before an endpoint can enter the argument. Fix $T < T_*$. Classical smoothness on the compact interval $[0, T]$ gives, for every finite n and q ,

$$\sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^q} < \infty.$$

Consequently,

$$\|\partial_t^n u\|_{L^2(0, T; H^q)}^2 \leq T \sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^q}^2 < \infty.$$

Every finite rectangle is therefore finite after its strict right endpoint has been fixed. In quantifiers, this says

$$\forall T < T_* \ \forall N, M \in \mathbb{N}_0 \ \exists C_{T, N, M} < \infty : \quad \mathfrak{R}_{N, M}(u; T) \leq C_{T, N, M}.$$

The constant may grow as $T \uparrow T_*$.

Whole-terminal control fixes the finite coordinate first and supplies one constant for every strict cutoff:

$$\forall N, M \in \mathbb{N}_0 \ \exists C_{N, M}(u_0, \nu, T_*) < \infty \ \forall T < T_* : \quad \mathfrak{R}_{N, M}(u; T) \leq C_{N, M}(u_0, \nu, T_*).$$

Equivalently,

$$\sup_{0 < T < T_*} \mathfrak{R}_{N, M}(u; T) \leq C_{N, M}(u_0, \nu, T_*) < \infty \quad \text{for every finite } (N, M).$$

Each finite pair may have its own constant. The uniformity concerns the approach of that fixed rectangle to T_* .

The datum-generated compatible-rectangle theorem supplies this whole-terminal statement for every finite pressure-complete rectangle of the same history.